David Mekhtiev

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Education

University of Pennsylvania Philadelphia, PA
Master of Science in Engineering, Computer Science *January 2027 — Expected 2028*
- Incoming for the Spring 2027 semester with a focus on software systems (databases, networking, distributed systems). This is an **online** M.S.E program and does not interfere with full-time employment.

University of Michigan Ann Arbor, MI
Bachelor of Science in Engineering, Computer Science and Engineering *Aug 2021 — May 2025*
- Cumulative GPA: 3.84 / 4.0 | Dean's List, Engineering Scholarship of Honor

Work Experience

Hardware & Systems Engineer September 2025 — Present
Analog Devices *Wilmington, MA*
- Created and maintain a GUI tool that automates several workflows for embedded IC design / evaluation. Implemented full-stack functionality using `React + TypeScript` for the UI, a `Python` FastAPI CRUD backend, and a RESTful client–server architecture to distribute jobs and manage dependencies across an internal Linux compute cluster.

Hardware & Systems Engineer Intern May 2025 — Aug 2025
Analog Devices *Wilmington, MA*
- Developed a custom machine learning model, data generation/augmentation and EDA infrastructure, as well as model training/evaluation scripts to provide performance and power estimates of a proprietary hardware subsystem.
- This DSP + NPU subsystem is currently utilized in several product lines and deals with audio processing and low power machine learning workflows. These efforts significantly reduced the time burden of gathering performance and power data.

TA for EECS 483 (Compiler Design) Jan 2025 — May 2025
University of Michigan College of Engineering *Ann Arbor, MI*
- Worked with Professor Max New to provide office hours, responses to Piazza questions, a discussion section, exam design/grading, and general support for course logistics during the Winter 2025 semester.
- Earned a 4.9/5 average rating from students across two evaluations, highlighting strong preparation, subject matter knowledge, and engaging weekly discussion sessions.

Hardware & Systems Engineer Intern May 2024 — Aug 2024
Analog Devices *Wilmington, MA*

Projects

Sharded Paxos Key/Value Service Nov 2024 — Dec 2024
- I implemented a distributed sharded key/value store with fault tolerance and update consistency using the Paxos algorithm. The system partitions keys across multiple replica groups, each managing a subset of shards, with a central shard master managing shard assignments and reconfigurations. The shard master exposed a very simple client API: `Join()`, `Leave()`, `Move()`, and `Query()` RPCs.

Parameterizable Superscalar Out-of-order Processor Jan 2024 — April 2024
- I designed and implemented a fully synthesizable 32-bit `RISC-V` out-of-order core in `SystemVerilog` that utilized a MIPS R10K inspired microarchitecture. Advanced features included a custom GUI debugger made with `React` that scraped `.fsdb` dumps to display signals every clock cycle, a tournament predictor, load-store queue, instruction prefetching, and multi-ported non-blocking caches.

Virtual Memory Pager Oct 2023 — Nov 2023
- I created a simulator of the virtual memory pager portion of a Unix-like operating system kernel used to manage the virtual address space of application processes. The pager was written in `C++` and implemented system calls like `fork()` to create, copy, and destroy address spaces, allocate more space in existing ones, and switch between them.

Skills

- **Programming Languages**: `C`, `C++`, `CUDA`, `Golang`, `JavaScript`, `Python`, `Rust`, `SQL`, `SystemVerilog`, `TypeScript`
- **Tools**: Docker, GDB, Git, LLDB, PostgreSQL, Valgrind, WSL
- **Libraries/Frameworks**: Boost, Electron, FlatBuffers, Matplotlib, NumPy, React, TensorFlow